AI-Driven Export Market Decision Support System for Indian Exporters Using Public Trade and Macroeconomic Data

Interim Report

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GitHub Repository

https://github.com/bharathipavithra10-cmd/AI-Export-Market-Decision-Support

Introduction

Background and Context

International trade is a major contributor to India's economic growth, yet selecting suitable export markets remains a complex decision requiring the simultaneous evaluation of market demand, historical growth, supplier competition, and macroeconomic conditions. This challenge is particularly significant for micro, small, and medium enterprises (MSMEs), which often have limited access to integrated market-intelligence systems and advanced analytical resources.

Public sources such as TradeStat, UN Comtrade, and World Bank Open Data provide valuable information, but the datasets are maintained separately and differ in structure, naming conventions, units, coverage, and reporting periods. Considerable cleaning and integration are therefore required before they can support a consistent product-country-year analysis. TradeStat provides India-specific exports by commodity and destination; UN Comtrade provides broader international trade flows; and World Bank indicators describe the economic environment of destination countries.

The project develops an AI-driven Export Market Decision Support System that combines public trade and macroeconomic data with descriptive analysis, statistical methods, machine learning, and business-intelligence visualization. The intended outcome is a transparent framework that helps Indian exporters identify promising markets, understand the reasons behind rankings, and distinguish between large established destinations and emerging markets with favorable growth conditions.

Problem Statement

Indian exporters currently rely on fragmented trade information, historical reports, and manual market analysis. Existing public portals provide descriptive statistics, but they do not jointly evaluate destination demand, India's export performance, supplier competition, macroeconomic conditions, and future potential in one workflow. Consequently, exporters may focus on large but saturated markets or overlook smaller markets with stronger growth opportunities. This study addresses the gap by creating an integrated, data-driven decision-support workflow using publicly available sources.

Purpose of the Study

The purpose of the study is to develop and evaluate an AI-driven decision-support system that assists Indian exporters in identifying and prioritizing international markets. The study analyzes export demand and historical trends, examines macroeconomic and trade-related drivers, plans predictive models for future export opportunities, and ultimately aims to rank destination markets using demand, competition, market share, macroeconomic conditions, and predicted growth.

Table 1

Interim Project Status (Progress Snapshot)

Status	Work items
Completed	Repository setup; TradeStat, World Bank and UN Comtrade data loading; data cleaning and integration; feature engineering; exploratory data analysis; missing-value analysis; multicollinearity

Status	Work items
	assessment (VIF); RQ1 statistical analysis (Kruskal–Wallis test); RQ2 multiple regression analysis; RQ4 predictive modelling using Linear Regression, Random Forest and XGBoost.
In progress	RQ3 supplier competition analysis (pending correction of UN Comtrade extraction); Export Market Attractiveness Score design and dashboard development.
Pending	Final RQ3 analysis, attractiveness scoring framework, dashboard deployment, business recommendations, model refinement and final report preparation.

Scope and Objectives

The initial synopsis proposed evaluation of at least 20 product categories over approximately ten years, subject to data availability. At the interim stage, the implemented notebook contains 29 TradeStat workbooks representing five HS chapters (27, 30, 71, 84, and 85) over the available annual periods. The current unit of analysis is the product-country-year observation. The report therefore distinguishes the initial target scope from the currently available analytical scope and does not overstate the completed coverage.

Objectives

1. Analyze current export demand and historical market trends for selected Indian export categories across international destinations.
2. Evaluate the association of macroeconomic and trade-related variables with export performance.

3. Develop and evaluate statistical and machine-learning models for future export demand and growth.
4. Build a transparent export-market decision framework that integrates demand, competition, market characteristics, and predictive analytics.

Research Questions

RQ1: Which Indian products and destination markets currently demonstrate the highest export demand, and how have these patterns evolved over time?

RQ2: Which macroeconomic and trade-related factors significantly influence export demand across international markets?

RQ3: How do supplier competition and India's market share influence export opportunities across international markets?

RQ4: Which international markets represent the most attractive export opportunities based on demand, competition, market characteristics, and future growth potential?

Literature Survey

Literature Review Approach

The literature base was selected to cover four themes aligned with the research questions: international market selection, export diversification and product-space analysis, supplier concentration and market competition, and the use of machine learning for trade forecasting. Peer-reviewed journal articles, institutional methodologies, and recognized working papers were prioritized. Sources were retained when they offered a clear theoretical or methodological contribution to market screening, export opportunity assessment, competition measurement, or predictive modelling.

Table 2

Literature Review Summary

Author (Year)	Domain/context	Method	Key finding	Relevance
Andersen & Strandskov (1998)	International market selection	Conceptual/ cognitive mapping	Market choice depends on how decision makers structure multiple market signals.	Supports transparent multi-factor decision design.
Papadopoulos et al. (2002)	Market screening	Trade-off model	Market attractiveness requires balancing opportunity and risk dimensions.	Supports composite ranking rather than single-indicator selection.
Gaston-Breton & Martin Martin (2011)	Market selection and segmentation	Two-stage model	Initial country screening should precede detailed segmentation.	Supports staged workflow from screening to recommendation.
Brenton & Newfarmer (2007)	Export diversification	Cross-country trade analysis	Export growth reflects both intensive and extensive margins.	Supports analysis of established and emerging markets.
Hidalgo et al. (2007)	Product space	Network analysis	Countries develop through related product capabilities.	Supports commodity-level differentiation.
Hausmann & Hidalgo (2011)	Economic complexity	Network structure	Product mix and capabilities shape economic output.	Provides theoretical basis for product-market structure.
Caldarelli et al. (2012)	Export networks	Network analysis	Export relationships reveal structural patterns not captured by totals alone.	Supports later competition/network extension.
Decreux & Spies (2016)	Export potential	ITC assessment methodology	Export opportunities combine supply, demand, and market-access considerations.	Directly informs attractiveness scoring.

Author (Year)	Domain/context	Method	Key finding	Relevance
Rhoades (1993)	Supplier concentration	HHI methodology	HHI summarizes market concentration using squared shares.	Supports planned RQ3 competition metric.
Batarseh et al. (2019)	Trade forecasting	Machine learning	ML can identify nonlinear patterns in international trade data.	Supports planned comparison of predictive models.
El Kefi (2021)	International market selection	Data envelopment analysis	Multi-criteria efficiency techniques can prioritize export markets.	Offers benchmark for ranking design.
Silva et al. (2024)	Trade networks and forecasting	Machine learning/network features	Network structure may improve economic forecasting.	Supports future feature engineering.

Across the literature, international market selection is consistently treated as a multi-criteria decision rather than a simple ranking by market size. This supports the project's decision to combine export values, growth, macroeconomic variables, competition measures, and forecast outputs. A second theme is the importance of product-level heterogeneity: a country attractive for one HS chapter may not be attractive for another. Accordingly, the notebook retains commodity identifiers and calculates country and commodity summaries separately.

A third theme concerns competition and diversification. Export opportunity is not equivalent to high import demand; supplier concentration, India's current share, and market accessibility can materially change the attractiveness of a destination. These variables remain pending because the current UN Comtrade filtering produced an empty India subset. The limitation is documented rather than concealed, and RQ3 will be completed after the Comtrade extraction logic is corrected.

Finally, the forecasting literature supports comparison of interpretable baselines with nonlinear machine-learning models. The final project plans to compare linear regression, random forest regression, and XGBoost using out-of-time validation and error metrics such as MAE and RMSE.At the interim stage, Multiple Linear Regression, Random Forest, and XGBoost models have been developed and evaluated. Their comparative performance is presented in the Preliminary Model Performance section.

Data Description

Table 3

Data Sources and Access

Source	Notebook input	Purpose	Current status
TradeStat, Department of Commerce, Government of India	29 Excel files	India's exports by destination, commodity, current/previous value, and growth	Loaded and cleaned; 6,040 rows.
UN Comtrade	1 Excel file; 100,000 rows and 47 original columns	Destination imports, quantities, weights, partner information, and competition inputs	Loaded, reduced to 10 fields; India reporter filter returned zero rows and requires correction.
World Bank Open Data	1 Excel workbook; 2,125 initial rows	GDP, GDP growth, GDP per capita, population, inflation, exports/imports/trade as % of GDP	Cleaned, reshaped, pivoted, and merged by country-year.

Dataset Overview

The TradeStat loader read 29 commodity-year workbooks and combined them into 6,040 records with 10 source variables. The World Bank workbook initially contained 2,125 rows and 10 columns; after empty-row removal, 2,122 records remained. Reshaping the annual columns from 2020 through 2025 produced 12,732 country-indicator-year records, which were pivoted into a country-year table. Following country-name standardization and merging, the final EDA dataset contains 6,040 rows and 21 columns.

Table 4

Dataset Overview

Attribute	Interim value
Rows	6,040
Columns used in EDA	21
Unit of analysis	Commodity-country-year
TradeStat files	29
UN Comtrade files	1
World Bank files	1
Years represented in TradeStat summaries	2021-2026
HS chapters currently represented	27, 30, 71, 84, 85
Primary outcome	Export_Current / Export_Value
RQ2 predictors	GDP, GDP growth, GDP per capita, population, inflation, trade as % of GDP

Table 5

Data Dictionary

Variable	Definition	Type	Source	Role/handling
Serial_No	Source row number	Object	TradeStat	Identifier only
Country	Destination market	Categorical	TradeStat	Grouping and merge key
Commodity	HS chapter code	Categorical	TradeStat	Product grouping
Year	Reporting year	Integer	TradeStat	Time key
Export_Previous	Previous-period export value	Continuous	TradeStat	Growth baseline
Export_Current	Current-period export value	Continuous	TradeStat	Primary demand outcome
Export_Growth	Reported export growth (%)	Continuous	TradeStat	Source growth measure
Exports_GDP	Exports of goods/services (% GDP)	Continuous	World Bank	Trade openness predictor
GDP	GDP in current US dollars	Continuous	World Bank	Economic size predictor
GDP_Growth	Annual GDP growth (%)	Continuous	World Bank	Economic momentum predictor

Variable	Definition	Type	Source	Role/handling
GDP_Per_Capita	GDP per capita (current US$)	Continuous	World Bank	Income/market-development predictor
Imports_GDP	Imports of goods/services (% GDP)	Continuous	World Bank	Import orientation predictor
Inflation	Consumer-price inflation (%)	Continuous	World Bank	Stability predictor
Population	Total population	Continuous	World Bank	Market-size predictor
Trade_GDP	Trade as % of GDP	Continuous	World Bank	Trade openness predictor
Import_Value	Comtrade trade value	Continuous	UN Comtrade	Currently fully missing due to filter issue
Import_Quantity	Comtrade quantity	Continuous	UN Comtrade	Currently fully missing
Import_Net_Weight	Comtrade net weight	Continuous	UN Comtrade	Currently fully missing
Export_Value	Copy of Export_Current	Continuous	Derived	Convenience outcome
Export_Growth_Rate	Calculated growth from current/previous	Continuous	Derived	Growth analysis; infinities later replaced
Market_Share	Row export value / annual export total	Continuous	Derived	Relative contribution within year

GitHub Data Availability Statement

Raw and processed data, notebooks, figures, and code are maintained in the project repository.

Repository:

https://github.com/bharathipavithra10-cmd/AI-Export-Market-Decision-Support

Raw data:

https://github.com/bharathipavithra10-cmd/AI-Export-Market-Decision-Support/tree/main/data/raw

Processed / Clean data:

https://github.com/bharathipavithra10-cmd/AI-Export-Market-Decision-Support/tree/main/data/processed

Cleaned source datasets:

https://github.com/bharathipavithra10-cmd/AI-Export-Market-Decision-Support/tree/main/data/cleaned

Code / Notebooks:

https://github.com/bharathipavithra10-cmd/AI-Export-Market-Decision-Support/tree/main/notebooks

Outputs (Figures & Tables):

https://github.com/bharathipavithra10-cmd/AI-Export-Market-Decision-Support/tree/main/outputs

Analysis

Sample Size Assessment

The minimum sample size for each research question was estimated based on the planned analytical method to ensure sufficient statistical power and reliable model estimation. The actual sample size differed across research questions because each analysis required a different subset of complete observations after data cleaning and variable selection. Table 6 compares the minimum recommended sample size with the actual sample used in each analysis.

Table 6

Sample Size Assessment for Each Research Question

Research Question	Analysis Method	Minimum Required Sample	Actual Sample Used	Adequacy
RQ1	Market comparison + Kruskal-Wallis test	237	5806	Adequate
RQ2	Multiple Linear Regression	109	2651	Adequate
RQ3	Supplier competition and market concentration analysis	85	Pending	Pending
RQ4	Predictive modelling	109	4985	Adequate

Data Cleaning and Integration

The TradeStat, World Bank, and UN Comtrade datasets were preprocessed and integrated to create a consistent analytical dataset for the subsequent analyses. Data cleaning included removing blank and duplicate records, standardizing numeric variables, harmonizing country names, handling missing values, engineering analytical features, and integrating the datasets using Country and Year as the common merge keys. These preprocessing steps improved data consistency and ensured compatibility across the three data sources. Table 7 summarizes the principal cleaning and integration activities undertaken during this study.

During preprocessing, the UN Comtrade dataset was filtered to obtain India-specific supplier information for the selected commodities. However, the current extraction returned no valid India records after filtering, resulting in missing import-related variables. This issue has been documented as an interim limitation and will be resolved before completing the supplier competition analysis for Research Question 3.

World Bank indicator data were transformed into a country-year format, standardized, and merged with the TradeStat dataset. Macroeconomic indicators were converted to numeric values, while unused variables with no analytical value were removed.

Outliers were assessed using descriptive statistics, histograms, and boxplots. The export value distribution exhibited substantial positive skewness, with a small number of high-value observations representing genuine international trade activity rather than data errors. Consequently, these observations were retained. A logarithmic transformation of the dependent variable was applied during regression modelling, while HC3 heteroskedasticity-robust standard errors were used to improve the reliability of statistical inference.

Following data cleaning, feature engineering, and dataset integration, separate analytical datasets were created for each research question based on the availability of the required variables. These datasets formed the basis for the exploratory analysis, hypothesis testing, regression modelling, and predictive modelling presented in the subsequent sections.

Table 7

Data Cleaning and Integration Log

Issue	**Variables affected**	**Detection**	**Treatment applied**	**Rationale / next action**
Formatting and numeric strings	TradeStat value/growth fields	dtype inspectioncan y and coercion	Removed commas; converted with errors="coerce"	Creates valid numerical variables while preserving invalid entries as missing.
Total/blank rows	TradeStat	Workbook inspection	Dropped blank rows and country rows containing “Total”	Avoids double counting.
Duplicate rows	TradeStat	duplicated().sum()	No duplicates found	No action required.
Country naming	All sources	Merge-key inspection	Trimmed and title-cased names	Improves matching, though a formal concordance is still required.
All-null quantities	TradeStat quantity fields	Missing-value analysis	Dropped four unused quantity variables	They provide no analytical information.

Issue	Variables affected	Detection	Treatment applied	Rationale / next action
Empty Comtrade India subset	Import value/quantity/weight	Filter output and 100% missingness	Retained as documented limitation; filtering/extraction must be corrected	Required for RQ3 and true destination-demand metrics.
Infinite calculated growth	Export_Growth_Rate	Descriptive statistics showed infinity	Replaced positive/negative infinity with missing	Occurs when previous exports equal zero.
Partial World Bank coverage	Eight macro variables	Missingness percentages	Retained for preliminary pairwise correlation; complete-case/imputation strategy pending	Avoids undocumented imputation at interim stage.

Table 8

Missing-Value Analysis

Variable	Missing count	Missing %
Import_Value	6,040	100.00
Import_Quantity	6,040	100.00
Import_Net_Weight	6,040	100.00
Exports_GDP	3,204	53.05
Imports_GDP	3,204	53.05
Trade_GDP	3,204	53.05
Inflation	3,016	49.93
GDP	2,793	46.24
GDP_Per_Capita	2,793	46.24
GDP_Growth	2,792	46.23
Population	2,655	43.96
Export_Growth_Rate	615	10.18
Export_Growth	473	7.83
Export_Current	234	3.87

The missingness pattern has two distinct causes. First, the Comtrade variables are structurally missing because the reporter filter produced an empty subset; this is a data-extraction problem rather than random item nonresponse. Second, World Bank coverage varies by country and year, with approximately 44%-53% missingness across the macroeconomic fields. Consequently, complete-case datasets were created separately for each research question, resulting in different analytical sample sizes.

Table 9

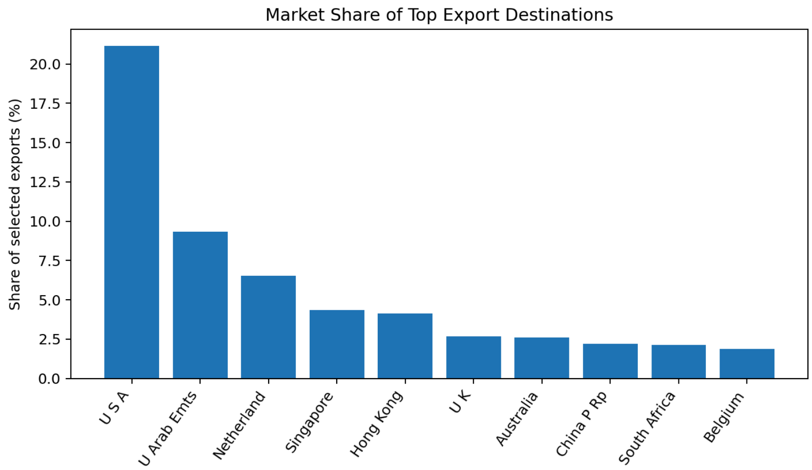
Descriptive Statistics and Distribution

Measure	Export_Current	GDP	GDP growth	GDP per capita	Population	Inflation
Count	5,806	3,247	3,248	3,247	3,385	3,024
Mean	184.43	3.19e11	4.10	21,640.78	2.39e7	9.45
Median	5.60	4.93e10	3.75	7,578.90	9.51e6	4.51
75th percentile	53.44	2.68e11	5.71	27,103.91	2.96e7	8.33
Maximum	25,682.64	5.23e12	63.33	288,001.57	2.86e8	359.09

Current export values are strongly right-skewed: the mean (184.43) is far above the median (5.60), and the maximum reaches 25,682.64. The notebook’s histogram and boxplot therefore indicate a large number of small product-country flows and a limited number of extremely large observations. This pattern supports later use of a logarithmic transformation or robust modelling strategy. Export growth also contains extreme observations when the previous value is very small or zero; the notebook subsequently replaces infinite growth values with missing values.

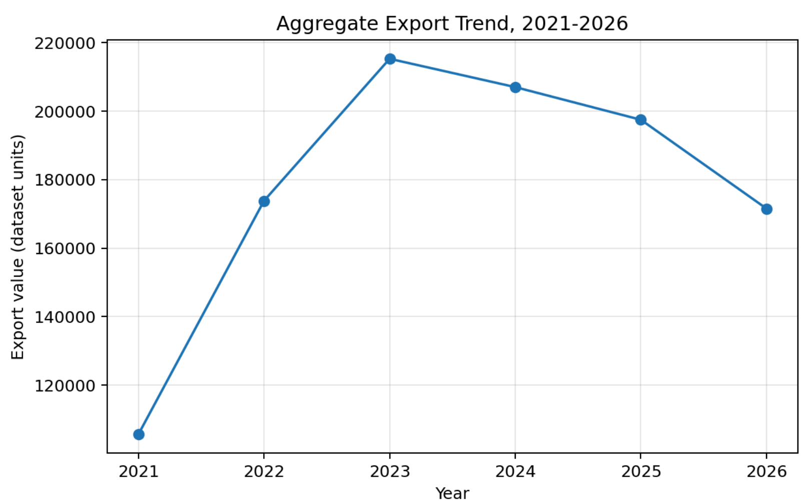
EDA Results (Interpretation Emphasis)

Figure 1



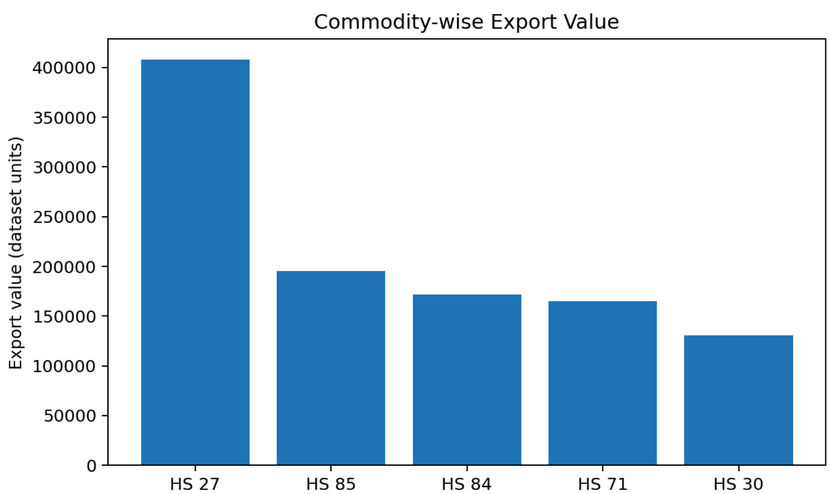
Insight: The United States accounted for 21.14% of export value, more than twice the share of the United Arab Emirates. This concentration indicates a need to balance established-market strength with diversification opportunities.

Figure 2



Insight: Figure 2 shows aggregate export value increasing from 105,623.29 in 2021 to 215,301.71 in 2023, followed by moderation to 207,030.96 in 2024, 197,492.25 in 2025, and 171,550.58 in 2026. The apparent 2026 decline must not be interpreted as a completed annual contraction until the reporting period is confirmed, because 2026 data may be partial. T

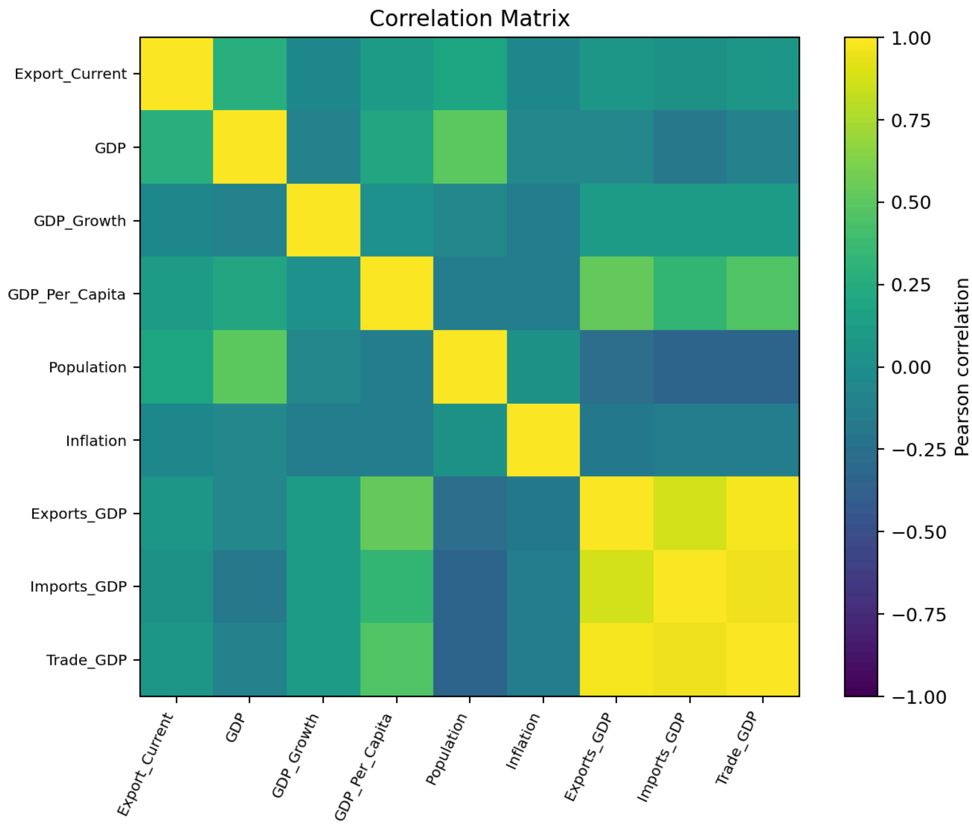
Figure 3



Insight: HS Chapter 27 recorded the highest export value throughout the study period, followed by HS Chapters 85, 84, 71, and 30. The variation across commodity groups supports commodity-specific export analysis.

Figure 4

Correlation Matrix for Export and Macroeconomic Variables



Insight: GDP showed the strongest positive correlation with export demand, followed by Population. Strong correlations among Exports (% GDP), Imports (% GDP), and Trade (% GDP) indicated multicollinearity, which was addressed using VIF analysis.

Table 10

EDA Insight Summary

Figure/Table	**What it shows**	**Key insight**	**Why it matters**	**Decision/next step**
Figure 1	Country shares	USA is dominant at 21.14%	Answers market-demand component of RQ1	Create product-specific destination rankings.
Figure 2	Annual export totals	Growth to 2023, then lower totals	Shows temporal evolution for RQ1	Verify whether 2026 is partial; use

Figure/Table	What it shows	Key insight	Why it matters	Decision/next step
				comparable annual periods.
Figure 3	Commodity totals	HS 27 dominates current sample	Demonstrates product heterogeneity	Avoid one ranking for all products.
Figure 4	Pairwise correlations	GDP has strongest export correlation; trade ratios are collinear	Preliminary evidence for RQ2	Use VIF and alternative model specifications.
Missing-value table	Data completeness	Comtrade variables 100% missing; macro variables partially missing	Constrains RQ2 and prevents RQ3	Correct extraction and define missing-data strategy.

Multicollinearity Assessment

Following the correlation analysis, multicollinearity among the predictor variables was assessed using the Variance Inflation Factor (VIF) before proceeding with regression modelling. Although the correlation matrix provided an initial indication of strong relationships among several macroeconomic indicators, VIF analysis was performed to quantify the extent of multicollinearity. The analysis was conducted using **2,688 complete observations** after excluding records with missing values in the selected predictor variables.

Table 11

Final Variance Inflation Factor (VIF) Results

Variable	Final VIF	Decision

GDP	1.93	Retained
GDP_Per_Capita	1.82	Retained
Population	1.67	Retained
Trade_GDP	1.55	Retained
GDP_Growth	1.06	Retained
Inflation	1.06	Retained

Modelling

Choice of Models with Justification

Three regression models were developed to evaluate export demand. Multiple Linear Regression was used as an interpretable baseline, while Random Forest and XGBoost were applied to capture nonlinear relationships and interactions within the structured trade dataset. The models were evaluated using an 80:20 train-test split, with Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 used to compare predictive performance. Random Forest achieved the strongest performance and was therefore selected as the preferred predictive model for the interim analysis.

Table 12

Choice of Models with Justification

Model	Purpose	Justification	Interim Status
Multiple Linear Regression	Estimate the influence of macroeconomic indicators on export demand	Provides interpretable coefficients, hypothesis tests, confidence intervals, and adjusted R^2 for statistical inference	Completed
Random Forest Regression	Predict export demand and identify key predictive features	Captures nonlinear relationships and feature interactions without requiring strict statistical assumptions	Completed
XGBoost Regression	Benchmark predictive performance against other	Well suited for structured tabular data and complex nonlinear relationships, often achieving high	Completed

	models	predictive accuracy	
Export Market Attractivenes s Score	Rank export markets based on multiple decision criteria	Combines normalized demand, growth, competition, macroeconomic indicators, and model predictions into a composite score	In Progress

Table 13

Features Included and Feature Engineering

Feature	**Original/e ngineered**	**Type**	**Reason for inclusion**	**Status**
Export_Current / Export_Value	Original/co py	Continuous	Current export demand outcome	Available
Export_Growth_ Rate	Engineered	Continuous	Measures period-over-period growth	Available; infinities treated
Export_Previous	Original	Continuous	Captures historical export performance and temporal persistence	Available ; Used in RQ4 models
Market_Share	Engineered	Continuous	Relative annual export contribution	Available, but not India share of destination imports
Commodity	Original	Categorical	Controls product heterogeneity	Available
Country	Original	Categorical	Controls destination heterogeneity	Available
Year	Original	Integer/time	Captures time pattern	Available
GDP	Original	Continuous	Market economic size	Available with missingness
GDP_Growth	Original	Continuous	Economic momentum	Available with missingness
GDP_Per_Capita	Original	Continuous	Market income/development	Available with missingness
Population	Original	Continuous	Market size	Available with missingness
Inflation	Original	Continuous	Economic stability	Available with missingness
Trade_GDP	Original	Continuous	Trade orientation/openness	Retained after VIF-based feature selection; Exports_GDP
Destination import value and HHI	Planned	Continuous	Demand and supplier competition	Unavailable until Comtrade correction

Table 14

Evaluation Metrics

Metric	Formula	Interpretation	Use
Growth rate	((Current - Previous) / Previous) x 100	Percentage change between periods	RQ1 trend analysis
Pearson correlation	$Cov(X,Y)/(\sigma X \sigma Y)$	Direction and strength of linear association	RQ2 – Correlation analysis
Adjusted R^2	$1 - [(1-R^2)(n-1)/(n-p-1)]$	Measures the proportion of variance explained after adjusting for the number of predictors	RQ2 Regression model evaluation
Variance Inflation Factor	$1/(1-Rj^2)$	Detects multicollinearity among predictors	RQ2 Multicollinearity assessment
MAE	$(1/n)\Sigma\|yi-\hat{y}i\|$	Average absolute prediction error	RQ4 model comparison
RMSE	$sqrt[(1/n)\Sigma(yi-\hat{y}i)^2]$	Penalizes larger errors	RQ4 model comparison
R^2	$1 - (SSres / SStot)$	Measures the proportion of variance explained by the predictive model	RQ4 – Predictive model comparison

Preliminary Results

Q1 Preliminary Findings

Analysis of 5,806 observations identified the United States as the largest export destination, accounting for 21.14% of total export value, followed by the United Arab Emirates (9.32%) and the Netherlands (6.52%). HS Chapter 27 recorded the highest export value, while HS Chapter 85 showed sustained growth over time. A Kruskal–Wallis test confirmed significant differences in export values across commodity groups, $H = 415.89$, $p < .001$. Therefore, $H0_1$ was rejected.

Table 15

Top Export Destination Markets (RQ1)

Rank	Destination	Share (%)
1	U S A	21.14
2	U Arab Emts	9.32
3	Netherland	6.52
4	Singapore	4.36
5	Hong Kong	4.12
6	U K	2.69
7	Australia	2.60
8	China P Rp	2.21
9	South Africa	2.12
10	Belgium	1.86

RQ2 Preliminary Findings

The RQ2 regression used 2,651 complete observations and six predictors retained after VIF-based feature selection. The overall model was statistically significant, $F = 107.48$, $p < .001$, and explained 23.3% of the variation in log-transformed export demand (adjusted $R^2 = .232$). GDP, Population, and GDP per Capita were significant positive predictors, while GDP Growth, Trade (% GDP), and Inflation were not significant at the 5% level. Therefore, $H0_2$ was rejected.

RQ3 Preliminary Findings

RQ3 remains in progress because the current UN Comtrade extraction did not produce usable India-specific supplier records. Consequently, India's destination market share, supplier count, and Herfindahl–Hirschman Index have not yet been calculated, and $H0_3$ has not been evaluated. The extraction logic will be corrected before the final report.

Table 16

Multiple Linear Regression Results

Predictor	Coefficient	p-value	Interpretation
GDP	0.592	<0.001	Significant positive influence
Population	0.492	<0.001	Significant positive influence
GDP per Capita	0.224	<0.001	Significant positive influence
GDP Growth	-0.063	0.055	Marginal negative relationship; not statistically significant
Trade (% GDP)	0.033	0.526	Not statistically significant
Inflation	-0.000	0.993	No statistically significant relationship

Preliminary Model Performance (RQ4)

Three predictive models were evaluated using an 80:20 train-test split consisting of 3,979 training observations and 1,006 testing observations (total sample = 4,985). Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Table 17

Preliminary Model Performance Comparison

Model	Feature Set	Validation Approach	MAE	RMSE	Key Takeaway
Multiple Linear Regression	Export_Previous, GDP, GDP Growth, GDP per Capita, Population, Inflation, Trade (% GDP), Country, Commodity, Year	80:20 Train-Test Split	19,185.39	579,601.44	Baseline model with poor predictive performance (R^2 = -365,776.36), indicating that linear regression could not capture the complex nonlinear relationships within the

					dataset..
Random Forest Regression	Same feature set	80:20 Train-Test Split	59.03	291.46	Best-performing model with the lowest prediction errors and highest predictive accuracy ($R^2 = 0.908$).
XGBoost Regression	Same feature set	80:20 Train-Test Split	65.11	363.92	Strong predictive performance, but slightly less accurate than Random Forest ($R^2 = 0.856$).

Interpretation: Random Forest produced the best predictive performance, followed by XGBoost, while Multiple Linear Regression performed poorly. These results indicate that nonlinear machine learning models are more suitable for predicting export demand than the baseline regression model. Therefore, $H0_4$ was rejected.

Interim Limitations and Risks

Despite the progress achieved in this interim phase, several limitations remain that will be addressed in the final report:

- The current analysis focuses on five major HS commodity chapters, rather than the full set of export product categories. Additional commodity groups will be incorporated during the final phase to improve the generalizability of the findings.
- Several World Bank macroeconomic indicators contain missing observations, reducing the usable sample size for the regression analysis from 5,806 to 2,651 observations.

- The UN Comtrade extraction currently requires further validation before supplier competition and market concentration measures can be computed for Research Question 3. Consequently, RQ3 remains incomplete in the interim report.
- Country names across multiple datasets required standardization. While harmonization procedures were applied, additional validation using ISO country codes will further improve data consistency.
- Export values exhibited substantial positive skewness and several extreme observations. These were retained because they represent genuine trade activity; instead, a logarithmic transformation and HC3 robust standard errors were applied to improve model reliability.
- The 2026 trade data may represent a partial reporting period and should therefore be interpreted with caution when comparing annual export trends.

Next Steps for the Final Report

The following activities will be completed during the final phase of the project:

1. Correct the UN Comtrade extraction and complete Research Question 3, including India's destination market share, supplier competition analysis, and the Herfindahl–Hirschman Index (HHI).
2. Develop the Export Market Attractiveness Score by combining historical export demand, macroeconomic indicators, supplier competition, market growth, and predictive model outputs.
3. Perform feature importance interpretation and sensitivity analysis to improve the explainability and robustness of the predictive models.

4. Build an interactive Power BI dashboard to visualize export opportunities, market rankings, and model predictions.
5. Validate the predictive modelling framework using additional robustness checks and finalize business recommendations for Indian exporters.
6. Complete the project documentation, including the repository README, data dictionary, and reproducibility guidelines.

References

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